

Sets

Notes By Pr. El Hadiq Zouhair

6.1 Definition

Definition 6.1 (Set). A *set* is an unordered collection of distinct objects, called its *elements* (or members). We write $a \in A$ when a is an element of A , and $a \notin A$ otherwise.

Sets can be described by *extension* (listing): $A = \{1, 2, 3, 5, 8\}$, or by *comprehension* (a defining property): $A = \{x \in \mathbb{N} \mid x \text{ is a Fibonacci number} \leq 8\}$. Order and repetition are irrelevant: $\{1, 2, 2, 3\} = \{3, 2, 1\}$.

Definition 6.2 (Equality). Two sets are equal when they have exactly the same elements: $A = B \iff \forall x (x \in A \leftrightarrow x \in B)$ (the *extensionality* principle).

6.2 Cardinality

Definition 6.3. If A contains exactly n distinct elements ($n \in \mathbb{N}$), A is *finite* and its *cardinality* is $|A| = n$. Otherwise A is *infinite*.

Example 6.4. $|\{a, b, c\}| = 3$; $|\emptyset| = 0$; $|\{\emptyset\}| = 1$ (a set containing one element, namely the empty set); $\mathbb{Z}$ is infinite.

6.3 Important sets

$$\mathbb{N} = \{0, 1, 2, \dots\}, \quad \mathbb{Z} = \{\dots, -1, 0, 1, \dots\}, \quad \mathbb{Q} = \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\}, \quad \mathbb{R}, \quad \mathbb{C}.$$

We write $\mathbb{Z}^+$ for the positive integers and $[a, b]$, (a, b) for real intervals. The chain $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ will be used constantly.

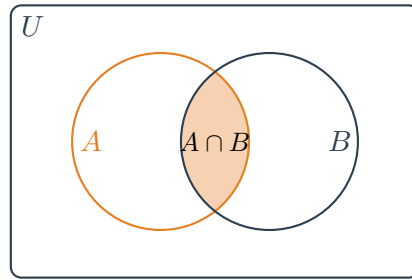
6.3.1 Countable and uncountable

Definition 6.5. A set is *countable* when it is finite or when its elements can be listed in an infinite sequence indexed by $\mathbb{N}$ (equivalently, when it has the same cardinality as $\mathbb{N}$, written $\aleph_0$). Otherwise it is *uncountable*.

$\mathbb{Z}$ is countable (list $0, 1, -1, 2, -2, \dots$) and so is $\mathbb{Q}$; Cantor's diagonal argument shows $\mathbb{R}$ is uncountable — there are strictly more reals than integers. Chapter 8 returns to cardinality via bijections.

6.4 Venn diagrams

A *Venn diagram* represents sets as regions (typically discs) inside a rectangle standing for the universal set U .



Venn diagrams guide intuition for the operations of Chapter 7, but a picture is not a proof: identities must be established by the element method or by membership tables.

6.5 Subsets

Definition 6.6. A is a *subset* of B , written $A \subseteq B$, when every element of A belongs to B : $\forall x (x \in A \rightarrow x \in B)$. If moreover $A \neq B$, A is a *proper subset*, written $A \subset B$.

Theorem 6.7. For every set A : $\emptyset \subseteq A$ and $A \subseteq A$. Moreover $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$ — the standard *double-inclusion* proof method.

Distinguish $\in$ from $\subseteq$: $1 \in \{1, 2\}$ but $1 \not\subseteq \{1, 2\}$; $\{1\} \subseteq \{1, 2\}$ and also $\{1\} \in \{\{1\}, 2\}$. Elements and subsets live at different levels.

6.6 The power set

Definition 6.8. The *power set* of A , written $\mathcal{P}(A)$, is the set of all subsets of A : $\mathcal{P}(A) = \{S \mid S \subseteq A\}$.

Example 6.9. $\mathcal{P}(\{a, b\}) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$; $\mathcal{P}(\emptyset) = \{\emptyset\}$.

Theorem 6.10. If $|A| = n$ then $|\mathcal{P}(A)| = 2^n$.

Proof. A subset is determined by deciding, for each of the n elements, whether it is in or out: n independent binary choices, hence 2^n subsets by the product rule. (An induction proof appears in Chapter 13.) $\square$

6.7 Partitions

Definition 6.11. A *partition* of a set A is a family of nonempty, pairwise disjoint subsets (*blocks*) whose union is A : every element of A lies in exactly one block.

Example 6.12. $\{\{1, 3, 5, \dots\}, \{0, 2, 4, \dots\}\}$ — odds and evens — partitions $\mathbb{N}$ into two blocks. The sets $\{1, 2\}, \{2, 3\}$ do *not* partition $\{1, 2, 3\}$: the blocks overlap.

Partitions are exactly the structures induced by equivalence relations (Chapter 8) — this correspondence is one of the central theorems of discrete mathematics.

6.8 Cartesian products

Definition 6.13. The *ordered pair* (a, b) is determined by its components *and their order*: $(a, b) = (c, d) \iff a = c \wedge b = d$. The *Cartesian product* of A and B is

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

More generally $A_1 \times \dots \times A_n$ consists of all n -tuples.

Example 6.14. $\{1, 2\} \times \{a, b, c\} = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$, and $|A \times B| = |A| |B|$. Note $A \times B \neq B \times A$ in general. $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$ is the coordinate plane.

6.9 Associations and datasets

Two structures built from sets pervade computing. An *association* (dictionary, map) is a finite set of key–value pairs $\{(k_1, v_1), \dots, (k_n, v_n)\}$ with distinct keys — formally a function from a finite key set (Chapter 8). A *dataset* (table) is a finite set or list of tuples drawn from a product $A_1 \times \dots \times A_n$: each row is a tuple, each column an attribute — precisely the data model of relational databases, built on the relations of Chapter 8.

6.10 The empty set

Definition 6.15. The *empty set* $\emptyset = \{\}$ is the unique set with no elements.

Every universally quantified statement about its elements is vacuously true; it is a subset of every set; and $|\emptyset| = 0$. Beware: $\emptyset \neq \{\emptyset\}$ — the latter has one element.

6.11 Summary

Sets are unordered collections of distinct elements, equal exactly when they have the same members. Key constructions: subsets and the subset order, the power set (2^n subsets for an n -element set), partitions (disjoint exhaustive blocks), and Cartesian products (ordered tuples). $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are countable; $\mathbb{R}$ is not. These objects are the raw material for all the structures to come.

Exercises

Exercise 6.1 EASY

List the elements of $A = \{x \in \mathbb{Z} \mid x^2 < 10\}$ and give $|A|$.

Exercise 6.2 EASY

True or false (justify briefly): (a) $\{1, 2, 3\} = \{3, 1, 2, 1\}$; (b) $\emptyset \in \{\emptyset\}$; (c) $\emptyset \subseteq \emptyset$; (d) $\{a\} \in \{a, \{a\}\}$ and $\{a\} \subseteq \{a, \{a\}\}$.

Exercise 6.3 EASY

Write out $\mathcal{P}(\{1, 2, 3\})$ and check its cardinality against Theorem 6.10.

Exercise 6.4 MEDIUM

Which of the following families are partitions of $\{1, 2, 3, 4, 5, 6\}$? (a) $\{\{1, 2\}, \{3, 4\}, \{5, 6\}\}$; (b) $\{\{1, 2, 3\}, \{3, 4, 5, 6\}\}$; (c) $\{\{1, 4\}, \{2, 5\}\}$; (d) $\{\{1\}, \{2, 3, 4, 5, 6\}, \emptyset\}$.

Exercise 6.5 MEDIUM

Let $A = \{0, 1\}$. Compute $A \times A$, $A \times A \times A$ and $\mathcal{P}(A) \times A$, giving the cardinality of each.

Exercise 6.6 MEDIUM

Prove that $A \subseteq B$ if and only if $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

Exercise 6.7 MEDIUM

Show that $\mathbb{Z}$ is countable by exhibiting an explicit enumeration, and give the formula of a bijection $f : \mathbb{N} \rightarrow \mathbb{Z}$ realising it.

Exercise 6.8 HARD

How many partitions does a 3-element set have? List them all for $\{a, b, c\}$. (The counts for growing n are the *Bell numbers*.)

Exercise 6.9 HARD

Prove that $(A \times B = B \times A)$ holds if and only if $A = B$, or $A = \emptyset$, or $B = \emptyset$.

Exercise 6.10 HARD

(Cantor, preview of Chapter 8.) Show that no function $f : A \rightarrow \mathcal{P}(A)$ is surjective, by considering $D = \{x \in A \mid x \notin f(x)\}$. Conclude that $|\mathcal{P}(A)| > |A|$ for every set A .

Solutions to the Exercises

Solution 6.1. $x^2 < 10$ for $x \in \{-3, -2, -1, 0, 1, 2, 3\}$, so A has these 7 elements: $|A| = 7$. ■

Solution 6.2. (a) True — repetition and order are irrelevant. (b) True — the set $\{\emptyset\}$ has $\emptyset$ as its single element. (c) True — every set is a subset of itself. (d) Both true: $\{a\}$ is listed as an element; and its only element a also belongs to $\{a, \{a\}\}$, giving the inclusion. ■

Solution 6.3. $\mathcal{P}(\{1, 2, 3\}) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$: $8 = 2^3$ subsets, as predicted. ■

Solution 6.4. (a) Yes: nonempty, disjoint, union is everything. (b) No: blocks overlap at 3. (c) No: union misses 3 and 6. (d) No: a partition may not contain the empty block. ■

Solution 6.5. $A \times A = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$, $|A \times A| = 4$. A^3 has $2^3 = 8$ triples (x, y, z) with $x, y, z \in \{0, 1\}$. $\mathcal{P}(A) = \{\emptyset, \{0\}, \{1\}, \{0, 1\}\}$ has 4 elements, so $\mathcal{P}(A) \times A$ has $4 \times 2 = 8$ pairs (S, a) . ■

Solution 6.6. $(\Rightarrow)$ Assume $A \subseteq B$ and let $S \in \mathcal{P}(A)$, i.e. $S \subseteq A$. Then $S \subseteq A \subseteq B$, so $S \in \mathcal{P}(B)$. $(\Leftarrow)$ Assume $\mathcal{P}(A) \subseteq \mathcal{P}(B)$. Since $A \in \mathcal{P}(A)$, we get $A \in \mathcal{P}(B)$, i.e. $A \subseteq B$. ■

Solution 6.7. Enumerate $0, 1, -1, 2, -2, 3, -3, \dots$. Formally define $f : \mathbb{N} \rightarrow \mathbb{Z}$ by $f(n) = (n+1)/2$ if n is odd and $f(n) = -n/2$ if n is even. Each integer appears exactly once (positive k at $n = 2k - 1$, nonpositive $-k$ at $n = 2k$), so f is a bijection and $\mathbb{Z}$ is countable. ■

Solution 6.8. Five partitions: $\{\{a\}, \{b\}, \{c\}\}$; $\{\{a, b\}, \{c\}\}$; $\{\{a, c\}, \{b\}\}$; $\{\{b, c\}, \{a\}\}$; $\{\{a, b, c\}\}$. The Bell numbers begin $1, 1, 2, 5, 15, 52, \dots$. ■

Solution 6.9. If $A = B$ or either is empty, both products are equal (both empty in the latter case). Conversely suppose $A \times B = B \times A$ with both nonempty. Take any $a \in A$; pick $b \in B$. Then $(a, b) \in A \times B = B \times A$, so $a \in B$; hence $A \subseteq B$. Symmetrically $B \subseteq A$, so $A = B$. ■

Solution 6.10. Suppose $f : A \rightarrow \mathcal{P}(A)$ and let $D = \{x \in A \mid x \notin f(x)\} \in \mathcal{P}(A)$. If f were surjective, $D = f(d)$ for some $d \in A$. Then $d \in D \iff d \notin f(d) = D$ — a contradiction. So no f is surjective. The injection $a \mapsto \{a\}$ shows $|A| \leq |\mathcal{P}(A)|$; absence of a surjection rules out equality, hence $|\mathcal{P}(A)| > |A|$. ■